UNIT-1 Design Thinking Introduction

Definition/Concept

Design Thinking is a human-centered problem-solving approach that combines creativity, empathy, and logic to create solutions that truly meet people's needs.

It's widely used in business, education, healthcare, and social innovation.

Why It Matters:

- Solves complex problems in innovative ways.
- Improves products, services, and experiences.
- Builds skills in creativity, collaboration, and adaptability.

Key Characteristics:
Human-Centered – Focuses on understanding the people you're designing for.
Creative – Encourages out-of-the-box ideas.
Iterative – Involves trying, testing, and improving solutions.
Collaborative – Uses diverse perspectives for richer results.

Why Design Thinking is Important

- Puts People First
- Ensures solutions are based on real human needs, not just assumptions.
- Increases user satisfaction and acceptance.
- Encourages Innovation
- Promotes creative, out-of-the-box thinking.
- Helps discover fresh solutions for old or complex problems.
- Solves Complex Problems
- Useful for "wicked problems" with no clear answers.
- Breaks challenges into manageable, testable steps.
- Reduces Risk of Failure
- Early prototypes and testing prevent costly mistakes.
- Feedback-driven improvements make solutions more effective.
- Works Across Industries
- Applicable in business, education, healthcare, government, and social projects.
- Useful for products, services, processes, and experiences.
- Builds Essential Skills
- Boosts creativity, empathy, collaboration, and adaptability.
- Prepares individuals and teams for real-world challenges.

History of Design Thinking

Period / Year	Key People / Organization	Milestone in Design Thinking History
1950s–1960s	John E. Arnold (Stanford), Herbert A. Simon	Early ideas of *creative problem-solving* and *design as a science* emerge. Herbert Simon's book *The Sciences of the Artificial* (1969) lays the foundation for a structured design process.
1970s	Robert McKim (Stanford)	Introduces *visual thinking* and *integrated problem-solving* approaches in engineering design education.
1970	Horst Rittel & Melvin Webber	Introduced the concept of "wicked problems" – complex, ill-defined problems without clear solutions (e.g., poverty, urban planning). They argued that such problems needed collaborative, iterative, and user-focused methods.
1980s	Rolf Faste (Stanford)	Expands McKim's work, formalizes "design thinking" as a way to apply designers' methods to problem-solving beyond product design.
1991	David Kelley & IDEO	IDEO was formed (merger of David Kelley Design, ID Two, Matrix Product Design) and popularized *human-centered design* in business and product development.
2005	Stanford d.school (Hasso Plattner Institute of Design)	Launches academic programs for teaching *Design Thinking* as a 5-stage process: Empathize, Define, Ideate, Prototype, Test.
2009 onwards	Tim Brown (CEO of IDEO)	Published *Change by Design*, bringing design thinking into mainstream business strategy.
2010s–Present	Global businesses, NGOs, governments	Design thinking is widely adopted in sectors like healthcare, education, IT, government policy, and social innovation.

Conventional Problem-Solving vs. Design Thinking

Aspect	Conventional Problem-Solving	Design Thinking
Approach	Linear and analytical	Iterative and creative
Focus	Problem-focused (fix what's broken)	Human-focused (understand people's needs first)
Process	Define → Analyze → Solve	Empathize → Define → Ideate → Prototype → Test
Basis	Logic, past data, expert opinion	Empathy, experimentation, user feedback
Outcome	One "best" solution chosen early	Multiple ideas tested, refined, and evolved
Risk Handling	Avoids failure by planning extensively	Accepts failure as learning through quick tests
Innovation Level	Often incremental	Often innovative and disruptive
Collaboration	Can be siloed by department/expertise	Multidisciplinary teamwork
Timeframe	Spends more time upfront on analysis	Moves quickly from idea to prototype

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In short:

- **Conventional problem-solving** = Fixing a problem using analysis and logic, often in a straight line.
- **Design Thinking** = Exploring a problem creatively with empathy, prototyping, and iteration until a user-centered solution emerges.

Examples of business uses of Design Thinking

Industry	Example	How Design Thinking Was Used	Outcome
Technology	**IBM**	Applied design thinking workshops to improve internal collaboration and create user-focused software interfaces.	Reduced product development cycle time by 33%.
Retail	**Nike**	Used empathy research to design *Nike Flyknit*, a lightweight, eco-friendly	Boosted sales and sustainability brand image.

Industry	Example	How Design Thinking Was Used	Outcome
		shoe tailored to athletes' needs.	
Hospitality	**Airbnb**	Conducted deep user interviews with hosts and guests to redesign the booking experience and trust-building features.	Transformed from near-bankruptcy to a multi-billion-dollar company.
Healthcare	**Mayo Clinic**	Used observation in the empathy stage to redesign patient waiting areas and care flow.	Improved patient satisfaction and reduced wait times.
FMCG	**Procter & Gamble (P&G)**	Applied design thinking to develop *Swiffer* after observing cleaning habits in homes.	Created a billion-dollar product line.
Automotive	**Ford**	Used rapid prototyping and user testing to create SYNC in-car connectivity systems.	Enhanced customer experience and brand loyalty.
Banking & Finance	**Bank of America**	Developed the "Keep the Change" program after studying customer saving habits.	Millions of new accounts and increased savings engagement.

TECHNICAL APPLICATIONS OF DESIGN THINKING

SOFTWARE DEVELOPMENT & IT

Redesigning a nospital's patient management software

• Empathy • Define • Ideate • Test

ENGINEERING & PRODUCT DESIGN

Developing an eco-friendly water purification device for rural india

• Understand • Prototype • Test

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Building an AI chatbot for banking services

• Empathy • Define • Prototype • Test

MECHANICAL & ELECTRICAL ENGINEERING

Redesigning EV charging stations

• Empathy • Ideate • Prototype • Test

HEALTHCARE TECHNOLOGY

Portable infant warmers (like the Embrace Warmer)

• Identify • Prototype • Test

AEROSPACE & DEFENSE

NASA applied design thinking in·lanning

Business Uses of Design Thinking

1. Product Innovation

Purpose: Create new products that truly meet customer needs.

Example: Apple used design thinking to develop the iPhone by focusing on user experience rather than just technical specs.

2. Service Design & Improvement

Purpose: Enhance customer service, reduce friction in processes.

Example: Airbnb redesigned its booking process by interviewing hosts and guests, improving trust and ease of use.

3**. Business Model Innovation**

Purpose: Rethink how a company creates, delivers, and captures value.

Example: Netflix moved from DVD rentals to streaming after understanding shifting customer behaviours and frustrations.

4. **Customer Experience (CX) Enhancement**

Purpose: Make every customer interaction more engaging and satisfying.

Example: Starbucks applied design thinking to create a more personalized in-store and digital ordering experience.

5**. Organizational Change & Culture Building**

Purpose: Improve internal collaboration, employee engagement, and problem-solving.

Example: IBM trained employees worldwide in Enterprise Design Thinking, fostering cross-team innovation.

6. Process Optimization

Purpose: Make internal workflows faster, cheaper, and more user-friendly for employees.

Example: Procter & Gamble redesigned supply chain processes to be more responsive to customer demand.

7. Social Impact & Sustainability

Purpose: Develop solutions for social, environmental, and community issues.

Example: Unilever uses design thinking to develop sustainable packaging solutions that reduce waste.

8. Digital Transformation
Purpose: Guide technology adoption based on real user needs.

Example: Bank of America used design thinking to create "Keep the Change," a digital savings tool that helped millions save automatically.

Variety Within the Design Thinking Discipline
1. Different Process Models
While the core idea remains similar (human-centered + iterative), organizations and educators adapt the steps:

Stanford d.school Model → Empathize → Define → Ideate → Prototype → Test
IDEO Model → Inspiration → Ideation → Implementation
Double Diamond Model (UK Design Council) → Discover → Define → Develop → Deliver
IBM Enterprise Design Thinking → Hills → Playbacks → Sponsor Users

Formation of a Design Thinking Team
Design thinking thrives on **diverse perspectives**, so team formation is deliberate:
1. Define the Goal
- Identify the problem or opportunity the team will address.
- Clarify the scope (product, service, policy, etc.).

2. Build a Multidisciplinary Team
Include members with varied expertise, such as:
- **Designers** – for creative ideation and visualization.
- **Engineers/Technologists** – for technical feasibility.
- **Business/Marketing Experts** – for market fit and strategy.
- **Researchers/Anthropologists** – for user insights.
- **End-Users/Stakeholders** – to provide real-world perspectives.

3. Assign Roles
- **Facilitator** – guides the design thinking process.
- **Research Lead** – manages user research and empathy work.
- **Idea Generator(s)** – pushes creative boundaries.
- **Prototyper(s)** – quickly turn ideas into tangible forms.
- **Tester(s)** – gather user feedback and evaluate solutions.

4. Set Team Norms
- Encourage open-mindedness.
- Accept failure as learning.
- Practice active listening.
- Promote equal voice in discussions.

Importance of a Design Thinking Team

1. Diversity Fuels Innovation

Different disciplines and viewpoints lead to more original, well-rounded solutions.

2. Human-Centered Outcomes

Teams work closely with users, ensuring solutions address real needs—not assumptions.

3. Faster Problem-Solving

Collaborative, iterative work cycles accelerate from idea to prototype to market.

4. Better Risk Management

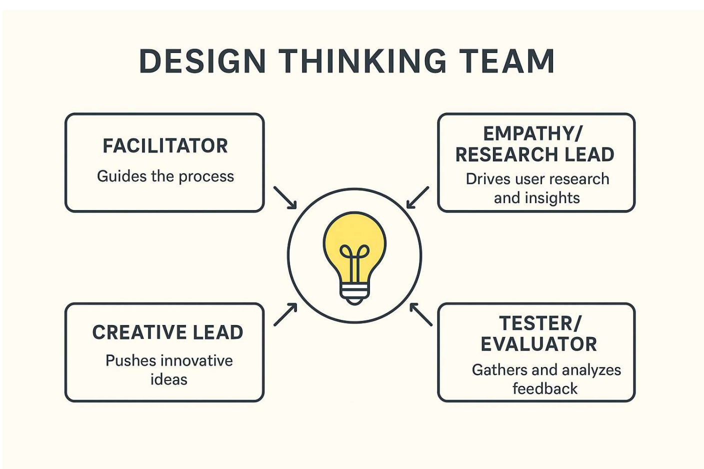
Testing and refining ideas early reduces costly failures later.

5. Organizational Impact

A strong design thinking team can foster a culture of innovation across the business.

💡 **Key takeaway:**

A design thinking team is like a **creative laboratory**—built on diverse skills, shared empathy, and rapid iteration. The right mix of people and roles can make the difference between an idea that fails quietly and one that transforms an industry.



Scenario: A school wants to improve the lunchtime experience for students

Design Thinking Team

Participants:

2 teachers

3 students from different grades

1 cafeteria manager

1 parent representative
1 design thinking facilitator

Scenario: A furniture manufacturing company wants to reduce assembly time for its modular office desks.
Design Thinking Team
Participants:
Production manager
2 assembly line workers
Quality control supervisor
Product designer
Supply chain manager
Design thinking facilitator

Scenario:
A software company is developing a mobile banking app and wants to improve the bill payment feature.
Design Thinking Team
Participants:
Product manager
2 UX/UI designers
1 backend developer
1 frontend developer
Customer support representative
Design thinking facilitator

Formation of Design Thinking Workshops & Meetings
Design thinking workshops and meetings are structured, collaborative sessions designed to help participants explore problems creatively and co-create solutions. Their formation usually follows these steps:

1. Define the Objective
Clearly state the purpose (e.g., understanding user needs, generating ideas, testing prototypes).

Link the workshop to a specific stage of the Design Thinking process (Empathize, Define, Ideate, Prototype, Test).

2. Select Participants
Include a diverse mix: designers, engineers, business experts, marketers, end-users, and decision-makers.
Diversity ensures different perspectives and prevents “groupthink.”

3. Prepare the Content & Tools
Gather research data, user personas, and any prior insights.

Arrange tools like sticky notes, markers, whiteboards, prototyping materials, or digital tools (e.g., Miro, FigJam).

4. Structure the Agenda
Keep activities time-boxed to maintain focus.

Example structure:
Warm-up (5–10 mins)
User insight review (15–20 mins)
Idea generation (30–60 mins)
Prototyping/sketching (60 mins)
Feedback & next steps (15–20 mins)

5. Assign Roles
Facilitator – keeps the session on track.
Scribe – documents ideas and decisions.
Timekeeper – ensures schedule is followed.
Participants actively contribute ideas and feedback.

6. Conduct & Capture Outcomes
Use visual thinking (drawings, diagrams, post-its) to make ideas tangible.
Document all outcomes for further analysis.